1 Introduction

The purpose of this lab is to introduce the concepts of integrator and differentiator circuits. The differentiator circuit simulates differentiation, where the integrator circuit is used to determine the integral of the input voltage. We are using a breadboard, power supply, signal generator, oscilloscope, and CircuitLab simulation enviorenment, we will simulate real-world integrator and differentiator circuits in this lab.

1.1 Preliminary Work

The analog integrator circuit shown in Figure 1 saturates quickly in practice due to the offset voltages that may exist in the circuit. Therefore, a practical integrator circuit with an additional resistor added in parallel with the capacitor is used to avoid saturation, as shown in Figure 1. Also the formula of circuit have been given on Figure 1.

$$V_0(t) = -\frac{1}{RC}\int_0^t V_i(\tau)d\tau + V_c(0)$$

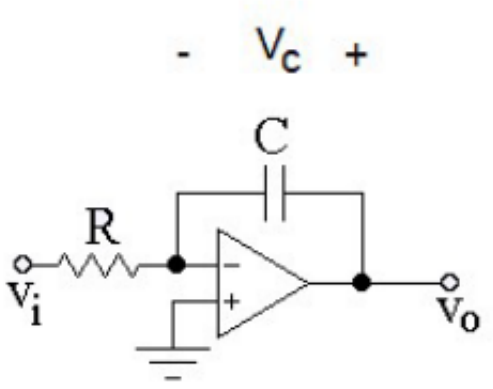


Figure 1: A first figure.

1. We simulated the practical integrator circuit in Figure 1 using CircuitLab and ran the simulation until 25 milliseconds.

2. We have built the circuit with R1 (5k), R2 (200k), C1 (10nF), V1 (-0.1) with LM741 OpAmp as shown in Figure 2.

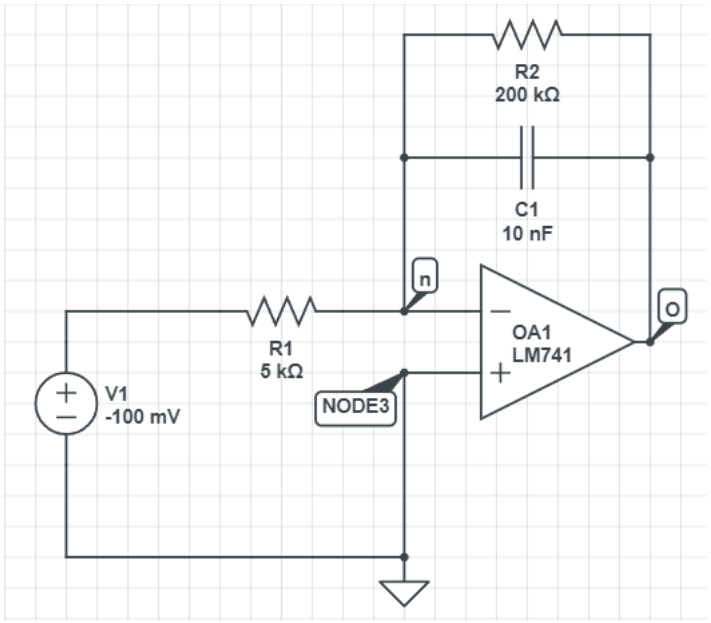


Figure 2: A first figure.

3. The simulation results showed that the output voltage approached saturation quickly and did not accurately represent the integral of the input voltage. This demonstrates the need for a practical integrator circuit with an additional resistor to avoid saturation and improve accuracy, as shown in Figure 3.

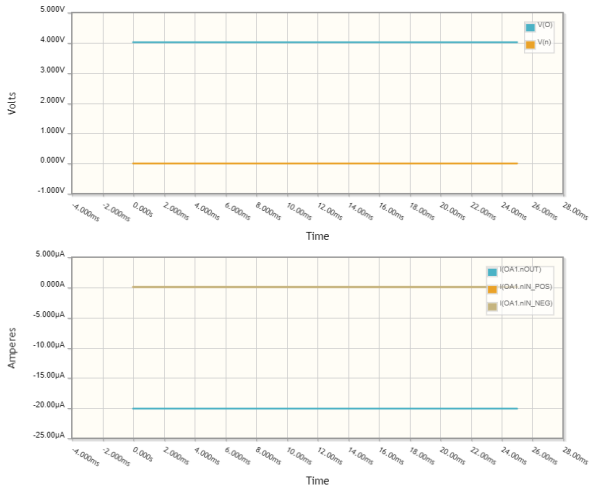


Figure 3: A first figure.

4. The differentiator circuit shown in Figure 3. A resistor is added in series with the capacitor in the practical differentiator circuit, as shown in Figure 4. Also the formula of circuit have been given on Figure 4.

$$Vout(t) = -RC\frac{d}{dt}Vin(t)$$

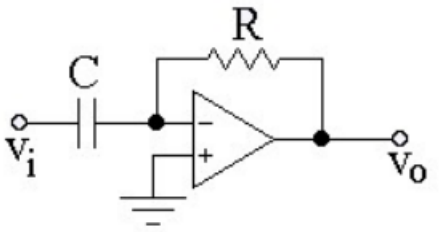


Figure 4: A first figure.

5. We have built the circuit with R1 (100), R2 (27k), C1 (22nF), V1 (0) with LM741 OpAmp as shown in Figure 5 below.

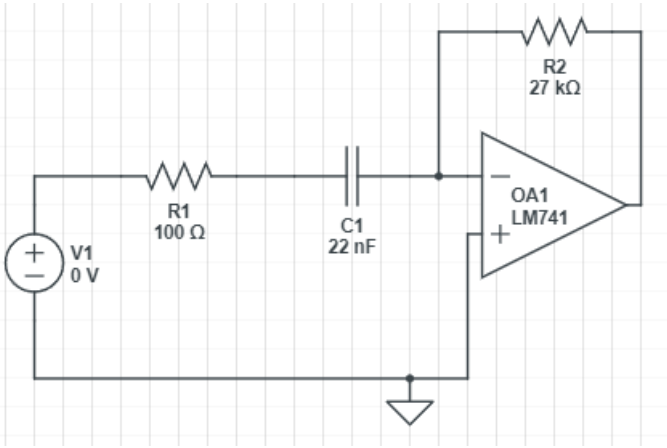


Figure 5: A first figure.

6. We simulated the differentiator circuit in Figure 5 using CircuitLab and ran the simulation until 2 milliseconds. The simulation results showed that the output voltage did not accurately represent the derivative of the input voltage.

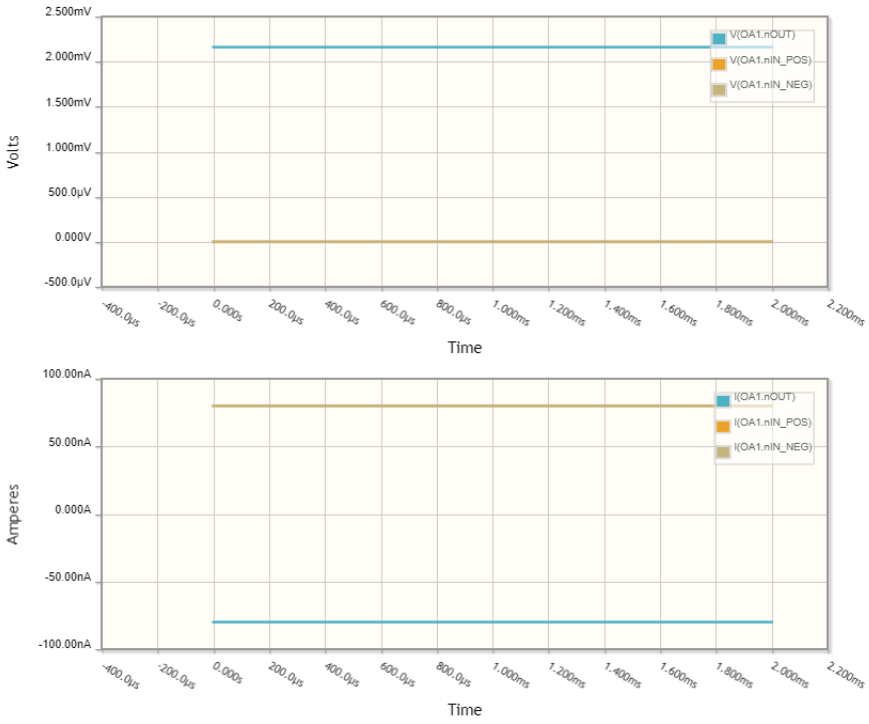


Figure 6: A first figure.

1.2 Lab Tasks

1.2.1 Lab Task 1

We constructed the integrator circuit shown in Figure 2 using a LM741 op-amp. We set up the power supplies (+15V and -15V) as shown in Figure 7. We set the signal generator to generate a 2 kHz, 0.1V amplitude square wave and connected it to the operational amplifier as shown in Figure 8.

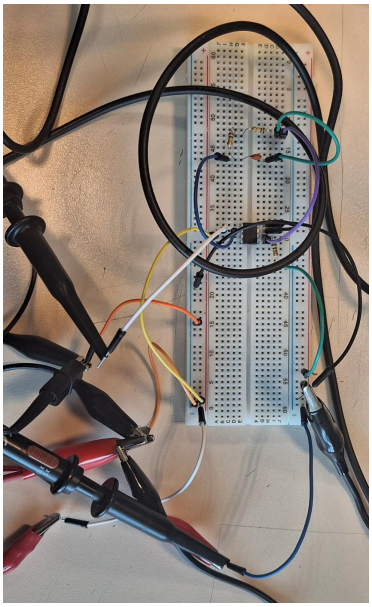


Figure 7: A first figure.

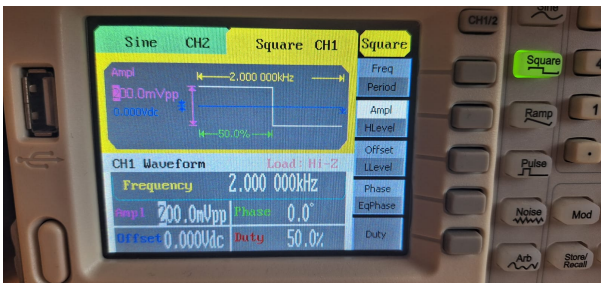


Figure 8: Wave generator generates square wave with 2kHz with 0.1V amplitude

1. We observed both the input and output wave forms on the oscilloscope screen (Figure 9) and measured the voltage output at a selected time to verify the operation of the integrator circuit formula on Figure 1. We repeated the experiment with a triangular wave and sinusoidal waves of 1 kHz and 10 kHz frequencies.

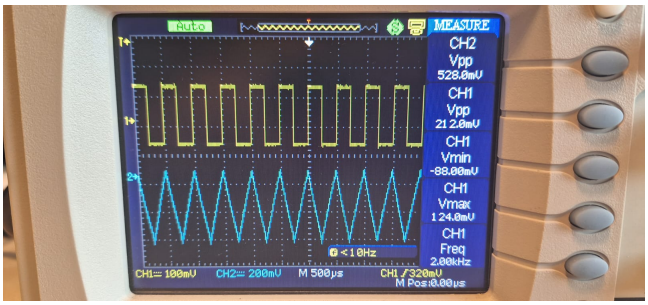


Figure 9: oscilloscope with given square input and as output triangular wave.

2. The results showed that the output voltage accurately represented the integral of the input voltage and did not saturate for any of the tested wave forms. The integrator circuit worked as expected and accurately calculated the integral of the input voltage as shown in Figure 9. In this experiment our expectation is obtaining triangular wave form when we provide square waves.

3. When we change the input waveform to a triangular wave (Figure 10) and repeat the previous step, with same values.

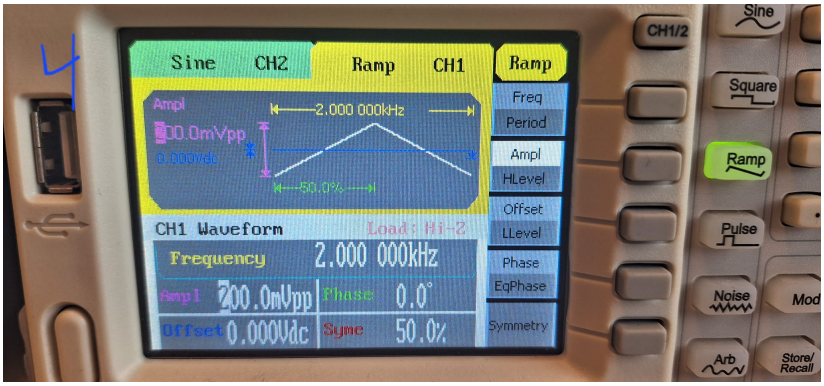


Figure 10: Wave generator generates triangular wave with 2kHz with 0.1V amplitude

4. Using the oscilloscope, we observed both the input and output wave forms on the screen. When we provide triangular waves we obtain sinusoidal waves as shown in Figure 11.

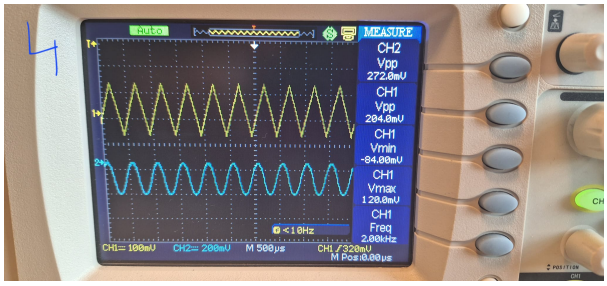


Figure 11: oscilloscope with given triangular input and as output sinusoidal wave.

5. When we change the input signal to 1 kHz sinusoidal wave (Figure 12) and repeated the 2nd step.

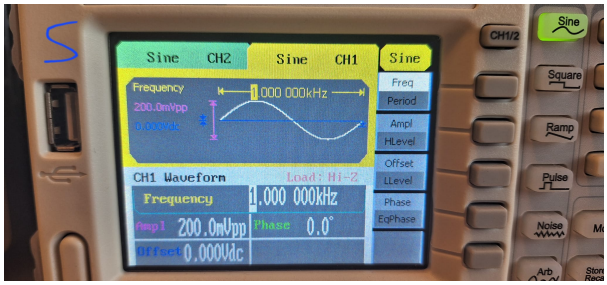


Figure 12: Wave generator generates sinusoidal wave with 1kHz with 0.1V amplitude.

6. Using the oscilloscope, we observed both the input and output wave forms on the screen. When we provide sinusoidal waves we obtain cosinusoidal waves as shown in Figure 13.

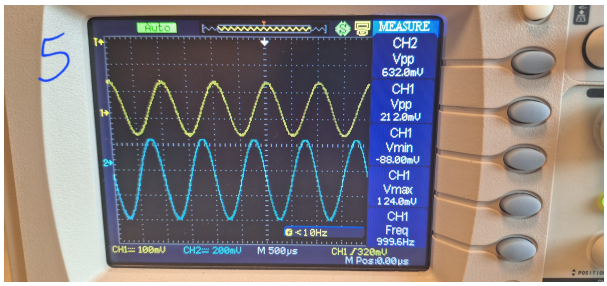


Figure 13: oscilloscope with given sinusoidal input and as output cosinusoidal wave.

7. When we change the input signal to a 10 kHz sinusoidal wave (Figure 14) and repeat 2nd step.

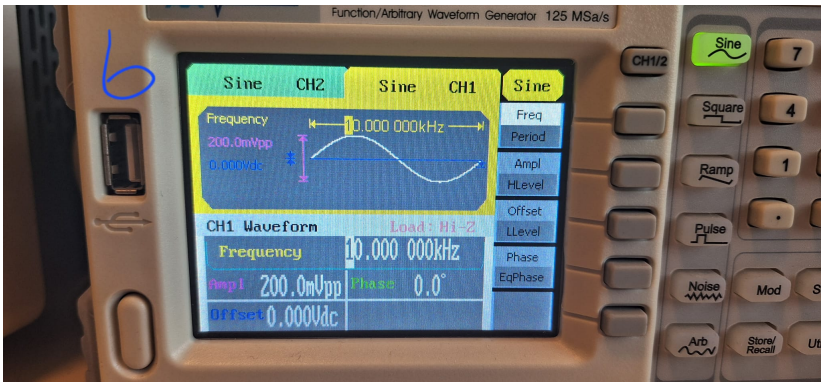


Figure 14: Wave generator generates sinusoidal wave with 10kHz with 0.1V amplitude.

8. Using the oscilloscope, we observed both the input and output wave forms on the screen. When we provide sinusoidal waves we obtain cosinusoidal waves as shown in Figure 15.

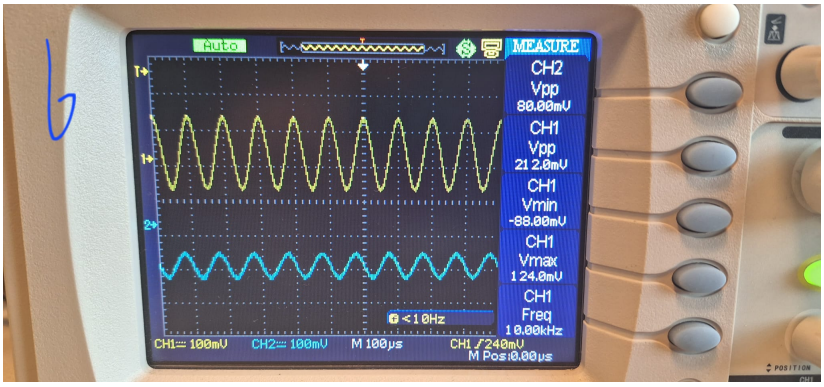


Figure 15: oscilloscope with given sinusoidal input and as output cosinusoidal wave.